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Check Valves: Wafer, Wafer-Lug, and Double Flanged Type

API STANDARD 594
FIFTH EDITION, NOVEMBER 1997



American Petroleum Institute

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Check Valves: Wafer, Wafer-Lug, and Double Flanged Type

Manufacturing, Distribution and Marketing Department

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FOREWORD

This standard is a purchase specification for wafer, wafer-lug and double flanged check valves designed for installation between Class 125 and 250 cast iron flanges as specified in ASME B16.1, between Class 150 and 300 ductile iron flanges as specified in ASME B16.42, between Class 150-2500 steel flanges as specified in ASME B16.5, and between Class 150-600 steel pipeline flanges as specified in MSS SP-44 or carbon steel flanges as specified in ASME B16.47.

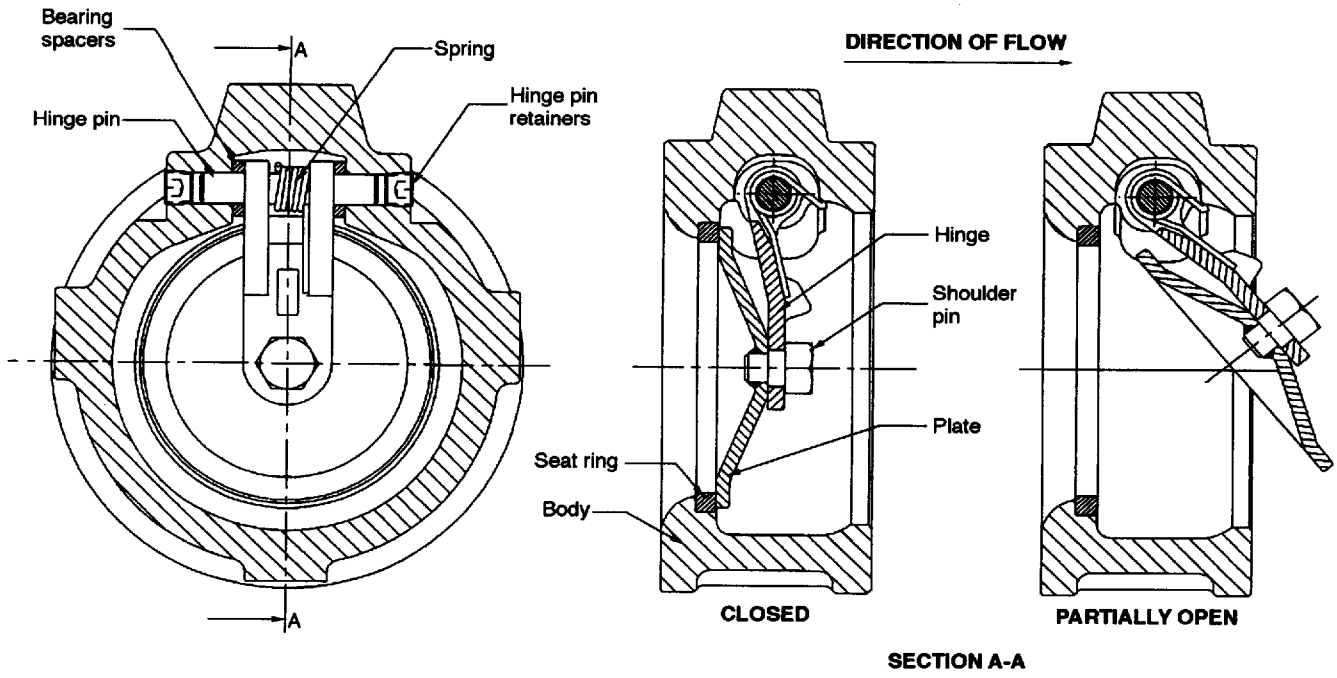
This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or incorporating sections thereof into another complete standard.

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Note: The optional configuration of a full flange or lug flange (similar to that shown in Figure 3) and of a double flanged type (similar to that of Figure 4) shall be the manufacturer's standard unless otherwise specified in the purchase order. All notes on Figures 3 and 4 apply.

Figure 1—Typical Single-Plate Wafer Check Valve

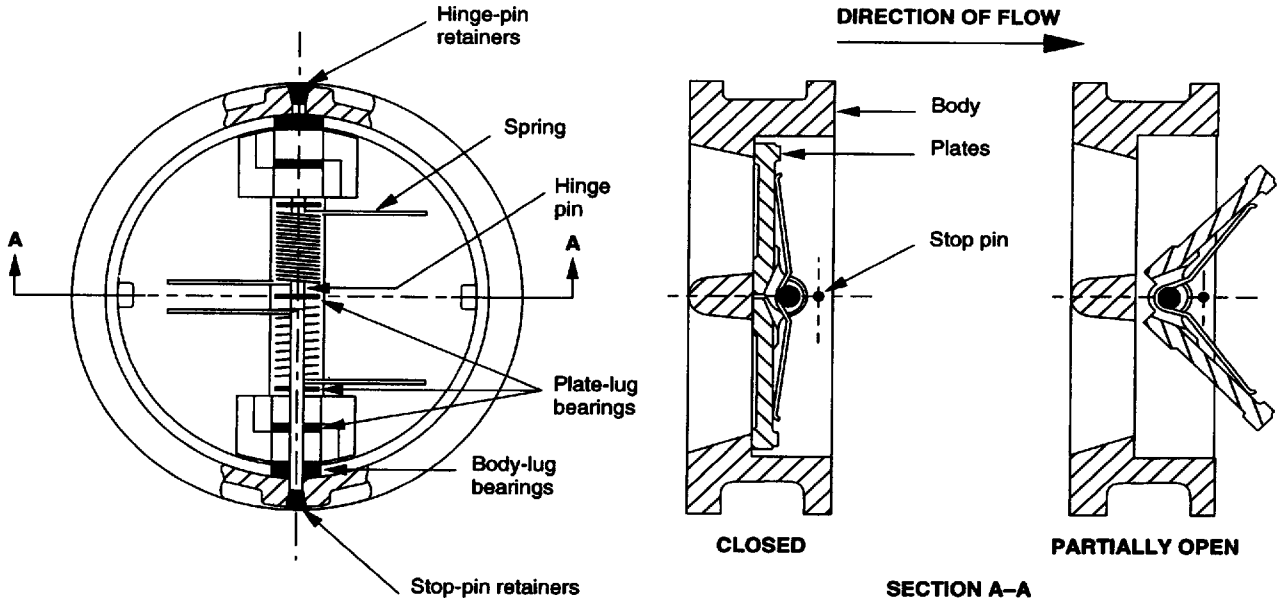


Figure 2—Typical Dual-Plate Wafer Check Valve

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IMPORTANT INFORMATION CONCERNING USE OF ASBESTOS OR ALTERNATIVE MATERIALS

Asbestos is specified or referenced for certain components of the equipment described in some API standards. It has been of extreme usefulness in minimizing fire hazards associated with petroleum processing. It has also been a universal sealing material, compatible with most refining fluid services.

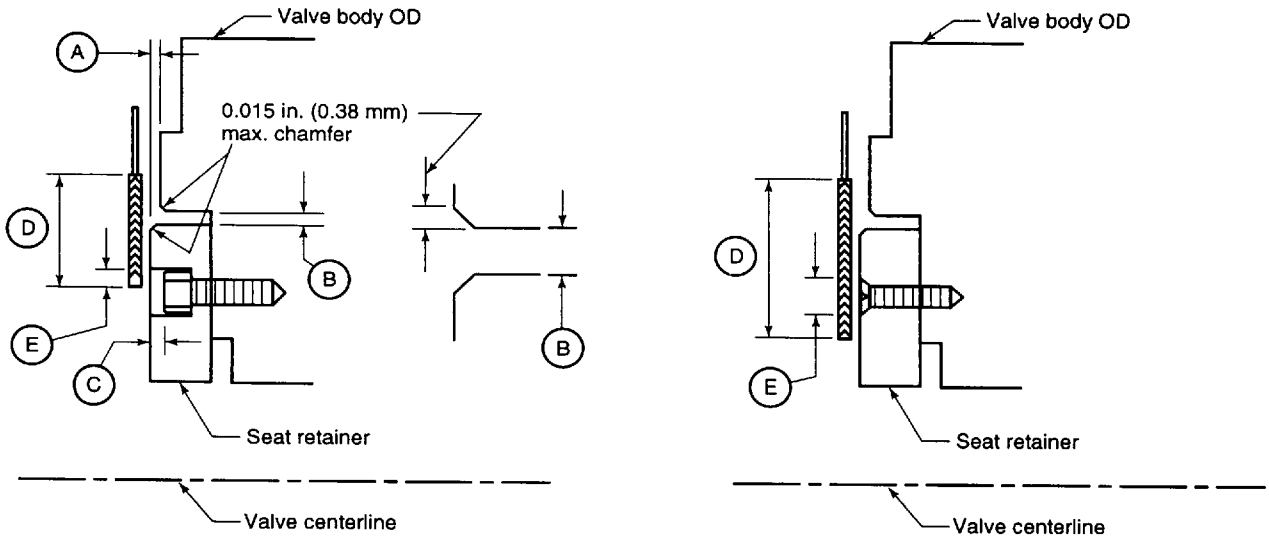
Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products (Sections 763.160-179).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

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Dimension	Definition	Range, inches	Range, mm
A	Protrusion of seat retainer plate above valve body face (after being compressed by mating flange). Negative value denotes insert below valve body face.	+0.010 to −0.010	+0.25 to −0.25
B	Radial width of annular gap between valve body and seat retainer plate (exclusive of chamfer).	0.030 max.	0.76 max.
C	Distance of screw head below face of seat retainer plate.	0 to 0.050	0 to 1.27
D	Width of sealing area of spiral-wound gasket for valve's size and rating.	-	-
E	Distance gasket sealing area overlaps fastener opening in face of seat retainer plate (may occur at the ID or OD of the gasket).	35 percent of D (max.)	35 percent of D (max.)

Figure 5—Limitations for Flange Face Interruptions that Fall Within the Gasket Seating Area

than the lower critical temperature of the base material. Cooling shall be in the furnace or in still air.

4.4 EXTERNAL BOLTS

4.4.1 Bolts 1 inch or smaller shall have coarse (UNC) threads. Those larger than 1 inch shall be of the eight-thread series (8 UN). Bolt threads shall be Class 2A, and nut threads shall be Class 2B. Threads shall conform to ASME B1.1.

4.4.2 For bolts 1 inch or smaller, threaded body-flange bolt holes shall be drilled and tapped according to the requirements for coarse thread series Class 2B in ASME B1.1. For bolts larger than 1 inch, such holes shall be drilled and tapped according to the requirements for eight-thread series Class 2B in ASME B1.1.

4.5 FLOW INDICATION

The valve body shall be furnished with a clearly visible cast, forged, machined-in, or die-stamped arrow to indicate the direction of flow through the valve.

4.6 GASKET SURFACE

Fasteners in the flange seating surface shall be recessed to or below the flange gasket level. Interruptions in the seating area of a centered ASME B16.20 Spiral Wound Gasket for valve sizes NPS 6 and larger shall not exceed the limitations given in Figure 5. The permissible surface interruptions on smaller size valves shall be as agreed between purchaser and manufacturer but shall not exceed 50% of the gasket seating width.

Note: The degree of interruption may affect the sealibility of a spiral-wound gasket.

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Table 1A—Minimum Body-Wall Thickness (Exclusive of Removable Seat Ring) by Class (Inches)

Valve Size (NPS)	Class								
	125	250	150	300	400	600	900	1500	2500
2	0.27	0.39	0.34	0.38	0.44	0.44	0.75	0.75	0.88
21/2	0.27	0.43	0.38	0.44	0.47	0.47	0.88	0.88	1.00
3	0.33	0.49	0.41	0.47	0.50	0.50	0.75	0.94	1.19
4	0.43	0.54	0.44	0.50	0.50	0.62	0.84	1.12	1.41
5	0.43	0.60	—	—	—	—	—	—	—
6	0.49	0.65	0.47	0.62	0.66	0.75	1.03	1.50	1.91
8	0.54	0.71	0.50	0.69	0.75	1.00	1.25	1.88	2.44
10	0.65	0.82	0.56	0.75	0.84	1.12	1.44	2.25	2.66
12	0.71	0.88	0.62	0.81	0.94	1.25	1.66	2.62	3.41
14	0.77	0.98	0.66	0.88	1.06	1.38	1.81	2.75	—
16	0.88	1.09	0.69	0.94	1.12	1.50	2.06	3.12	—
18	0.93	1.21	0.72	1.00	1.19	1.62	2.25	3.50	—
20	0.98	1.31	0.75	1.06	1.31	1.75	2.50	3.88	—
24	1.09	1.42	0.81	1.19	1.44	2.00	2.88	4.50	—
30	1.26	1.75	0.91	1.38	1.69	2.38	—	—	—
36	1.42	2.08	1.00	1.57	1.95	2.77	—	—	—
42	1.58	2.41	1.09	1.76	2.20	3.15	—	—	—
48	1.75	2.73	1.19	1.95	—	—	—	—	—
54	—	—	*	*	—	—	—	—	—
60	—	—	*	*	—	—	—	—	—

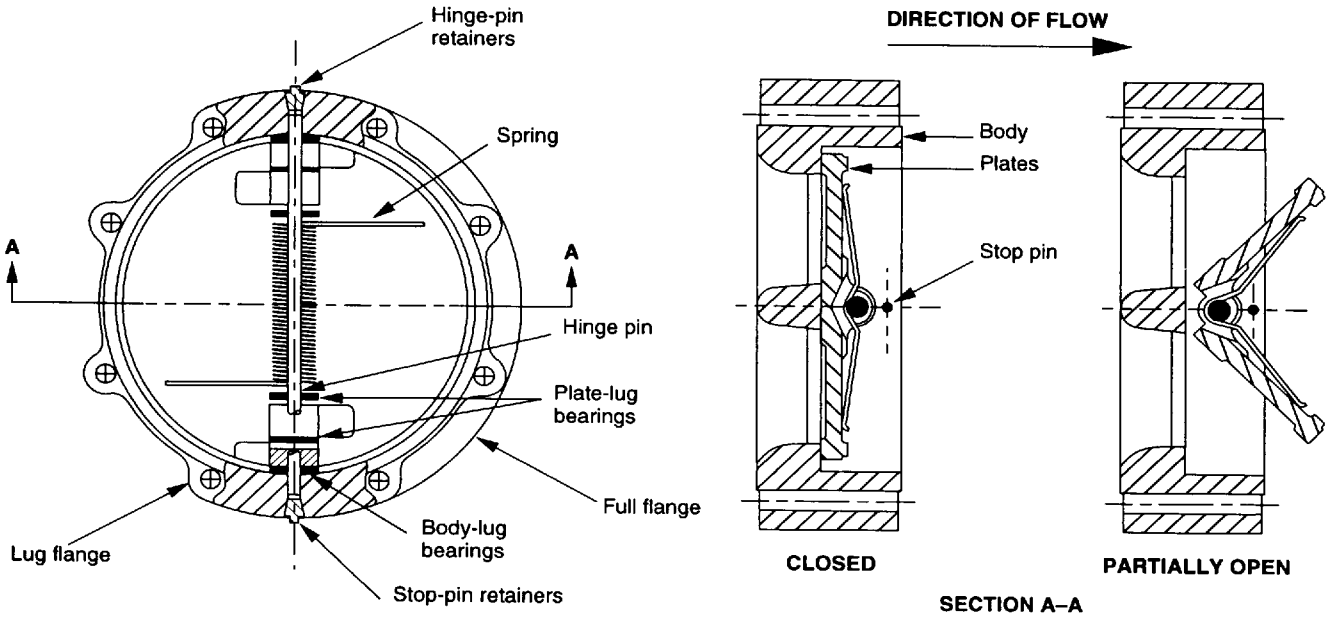
Note: The wall thickness shown for Classes 125 and 250 conform to those in ASME B16.1, except for NPS 36, 42, and 48, Class 250, which have been extrapolated. The wall thicknesses shown for Classes 150-2500 for sizes through NPS 24 conform to those in API Standard 600. The wall thicknesses for NPS 30-48 are extrapolations of the values in API Standard 600.
*Dimensional information to be agreed upon between purchaser and manufacturer.

Table 1B—Minimum Body-Wall Thickness (Exclusive of Removable Seat Ring) by Class (Millimeters)

Valve Size (NPS)	Class								
	125	250	150	300	400	600	900	1500	2500
2	7	10	9	10	11	11	19	19	22
21/2	7	11	10	11	12	12	22	22	25
3	9	12	10	12	13	13	19	24	30
4	11	14	11	13	13	16	21	29	36
5	11	15	—	—	—	—	—	—	—
6	12	17	12	16	17	19	26	38	48
8	14	18	13	17	19	25	32	48	62
10	17	21	14	19	21	29	37	57	67
12	18	22	16	21	24	32	42	67	87
14	19	25	17	22	27	35	46	70	—
16	22	28	17	24	29	38	52	79	—
18	24	31	18	25	30	41	57	89	—
20	25	33	19	27	33	44	64	98	—
24	28	36	21	30	36	51	73	114	—
30	32	45	23	35	43	60	—	—	—
36	36	53	25	40	50	70	—	—	—
42	40	61	28	45	56	80	—	—	—
48	45	69	30	50	—	—	—	—	—
54	—	—	*	*	—	—	—	—	—
60	—	—	*	*	—	—	—	—	—

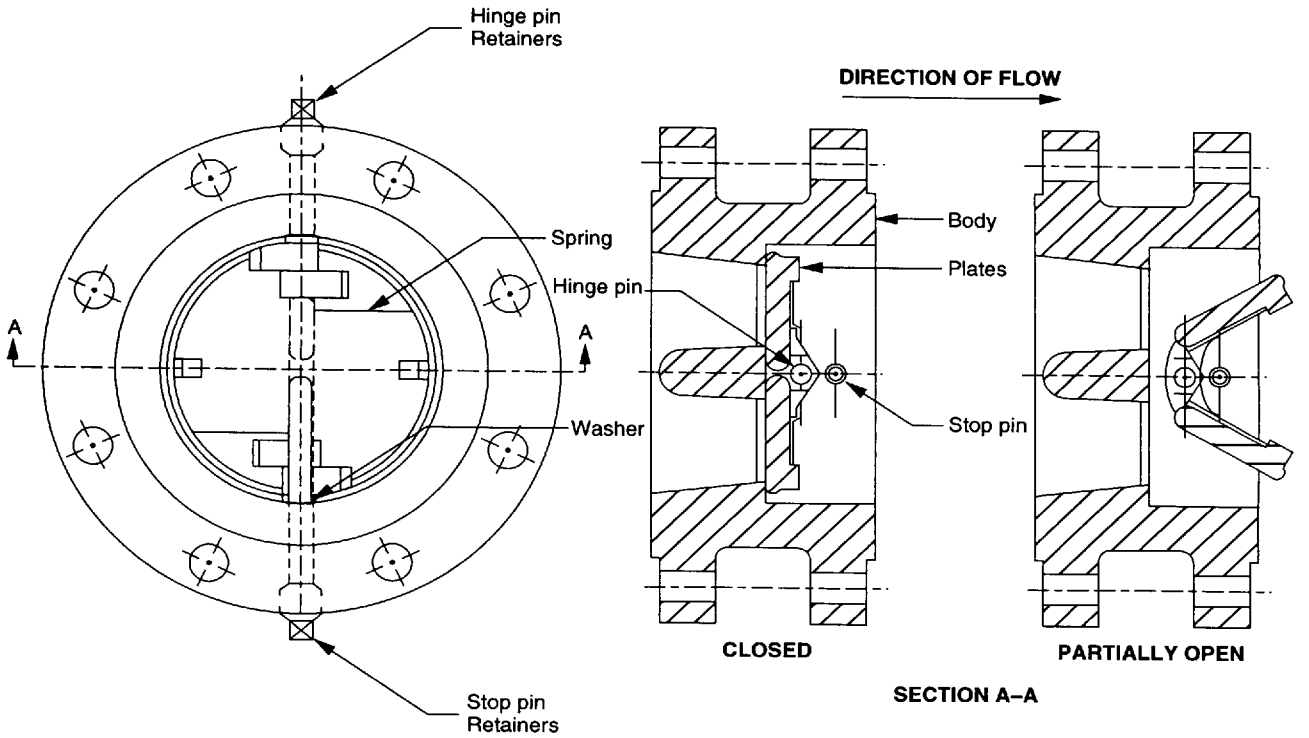
Note: The wall thicknesses shown for Classes 125 and 250 conform to those in ASME B16.1, except for NPS 36, 42, and 48, Class 250, which have been extrapolated. The wall thicknesses shown for Classes 150-2500 for sizes through NPS 24 conform to those API Standard 600. The wall thicknesses for NPS 30-48 are extrapolations of the values in API Standard 600.
*Dimensional information to be agreed upon between purchaser and manufacturer.

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Notes:
1. The optional configuration of a full flange or lug flange shall be the manufacturer's standard unless otherwise specified by the purchaser.
2. Unless otherwise specified in the purchase order, the bolt holes will be through-drilled.

Figure 3—Typical Dual-Plate Wafer-Lug Check Valve



Notes:
1. Insufficient space for nuts between flanges on some double flange valves may require other type valves be furnished.
2. Unless otherwise specified in the purchase order, the bolt holes will be through-drilled (see 4.1.11).

Figure 4—Typical Dual-Plate Double Flanged Check Valve

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Table 2A—Single- and Dual-Plate Valve Face-to-Face Dimensions by Class (Inches)

Valve Size (NPS)	Class								
	125	250	150	300	400	600	900	1500	2500
2	2.12	2.12	2.38	2.38	2.38	2.38	2.75	2.75	2.75
21/2	2.38	2.38	2.62	2.62	2.62	2.62	3.25	3.25	3.25
3	2.62	2.62	2.88	2.88	2.88	2.88	3.25	3.25	3.38
4	2.62	2.62	2.88	2.88	3.12	3.12	4.00	4.00	4.12
5	3.25	3.25	—	—	—	—	—	—	—
6	3.75	3.75	3.88	3.88	5.38	5.38	6.25	6.25	6.25
8	5.00	5.00	5.00	5.00	6.50	6.50	8.12	8.12	8.12
10	5.50	5.50	5.75	5.75	8.38	8.38	9.50	9.75	10.00
12	7.12	7.12	7.12	7.12	9.00	9.00	11.50	12.00	12.00
14	7.25	8.75	7.25	8.75	10.75	10.75	14.00	14.00	—
16	7.50	9.12	7.50	9.12	12.00	12.00	15.12	15.12	—
18	8.00	10.38	8.00	10.38	14.25	14.25	17.75	18.44	—
20	8.38	11.50	8.62	11.50	14.50	14.50	17.75	21.00	—
24	8.75	12.50	8.75	12.50	15.50	17.25	19.50	22.00	—
30	12.00	14.50	12.00	14.50	18.12	19.88	—	—	—
36	14.50	19.00	14.50	19.00	25.00	25.00	—	—	—
42	17.00	22.38	17.00	22.37	27.61	27.61	—	—	—
48	20.62	24.75	20.62	24.75	—	—	—	—	—
54	—	—	*	*	—	—	—	—	—
60	—	—	*	*	—	—	—	—	—

Note: The face-to-face tolerance shall be as specified in ASME B16.10 for sizes through NPS 24 and shall be ±0.125 inch (3 millimeters) for sizes larger than NPS 24.
*Dimensional information to be agreed between purchaser and manufacturer.

Table 2B—Single- and Dual-Plate Valve Face-to-Face Dimensions by Class (Millimeters)

Valve Size (NPS)	Class								
	125	250	150	300	400	600	900	1500	2500
2	54	54	60	60	60	60	70	70	70
21/2	60	60	67	67	67	67	83	83	83
3	67	67	73	73	73	73	83	83	86
4	67	67	73	73	79	79	102	102	105
5	83	83	—	—	—	—	—	—	—
6	95	95	98	98	136	136	159	159	159
8	127	127	127	127	165	165	206	206	206
10	140	140	146	146	213	213	241	248	254
12	181	181	181	181	229	229	292	305	305
14	184	222	184	222	273	273	356	356	—
16	191	232	191	232	305	305	384	384	—
18	203	264	203	264	362	362	451	468	—
20	213	292	219	292	368	368	451	533	—
24	222	318	222	318	394	438	495	559	—
30	305	368	305	368	460	505	—	—	—
36	368	483	368	483	635	635	—	—	—
42	432	568	432	568	701	701	—	—	—
48	524	629	524	629	—	—	—	—	—
54	—	—	*	*	—	—	—	—	—
60	—	—	*	*	—	—	—	—	—

Note: The face-to-face tolerance shall be as specified in ASME B16.10 for sizes through NPS 24 and shall be ±0.125 inch (3 millimeters) for sizes larger than NPS 24.
*Dimensional information to be agreed between purchaser and manufacturer.

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5 Material

5.1 BODY

The bodies of wafer, wafer-lug, and double flanged type check valves shall be made of a material conforming to a purchaser selected material specification listed in the applicable ASME standard as referenced in 3.2.

5.2 PLATES

Plates shall be made of a material whose corrosion resistance is greater than or equal to that of the valve body.

5.3 TRIM

5.3.1 The trim includes the following:

a. Body seating surfaces.
b. Plate seating surfaces.

5.3.2 Metallic seating surface material shall be manufacturer's standard which may be the same as the body material. Where specific trim is requested, it shall be as shown in Table 3. The typical specifications in Table 3 represent some acceptable grades. The resilient seal material, when required shall be specified by the purchaser and if located in the body there shall not be an overlay in the seat area unless specified in the purchase order.

5.4 INTERNAL WETTED PARTS

Internal wetted parts shall be suitable for the specified service conditions and their material will be manufacturers' standard unless otherwise specified in the purchase order. The term "wetted parts" shall include, but not be limited to, hinges, pins, bolts, bearings, and any other part in contact with the fluid medium other than the body, plates, trim, springs and pipe plugs.

5.5 BODY SEAT RINGS

If the body seat ring material is different from the seating-surface material, its corrosion resistance shall be greater than or equal to that of the valve-body material.

5.6 SPRINGS

Depending on the service, the material requirements for the spring may differ from those of the seat and other trim material and will be the manufacturer's standard unless otherwise specified in the purchase order.

5.7 PIPE PLUGS (PIN RETAINERS)

Any pipe plugs used shall be solid and shall have the same nominal chemical composition and material properties as does the valve body.

5.8 NAMEPLATE

The nameplate shall be 18Cr-8Ni steel or nickel alloy and shall be attached to the valve body by pins or welding. The pin material used for attachment shall be similar to the nameplate.

Table 3—Seating-Surface Nominal Trim Material

Trim No.	Nominal Trim	Material Type[a]	Typical Specification (Grade)		
			Cast	Forged	Welded
1	Type 410	11-13Cr	ASTM A 217 (CA15)	ASTM A 182 (F6)	AWS A5.9 (ER410)
2	Type 304	18Cr-8Ni	ASTM A 351 (CF8)	ASTM A 182 (F304)	AWS A5.9 (ER308)
5	Hard-faced	Co-Cr-A[b]	Manufacturer's standard	Manufacturer's standard	AWS A5.13 (E Co-Cr-A or R Co-Cr-A)
5A	Hard-faced	Ni-Cr[c]	Manufacturer's standard	Manufacturer's standard	Manufacturer's standard
9	Monel	Ni-Cu alloy	ASTM A 494 (M-35-1)	ASTM B 564 (UNS N04400)	Manufacturer's standard
10	Type 316	18Cr-8Ni-Mo	ASTM A 351 (CF8M)	ASTM A 182(F316)	AWS A5.9 (ER316)
13	Alloy 20	19Cr-29Ni	ASTM A 351 (CN7M)	ASTM B 473	AWS A5.9 (ER320)
15	Bronze	Bronze	Manufacturer's standard	Manufacturer's standard	—

[a]Resilient seating materials shall be specified by the purchaser.
[b]This classification includes but is not limited to, such trademarked materials as Stellite 6™, Stoody 6™, and Wallex 6™.
[c]These materials shall have manufacturer's standard hard facing with a maximum iron content of 25 percent.

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6 Inspection, Examination, Testing, and Repair

6.1 INSPECTION AND EXAMINATION

6.1.1 Each valve shall be visually examined by the manufacturer in accordance with API 598.

6.1.2 When inspection by the purchaser is specified by the purchase order, it shall be in accordance with API 598.

6.2 PRESSURE TESTS

Each valve shall be pressure tested in accordance with API Standard 598.

6.3 REPAIR OF DEFECTS

6.3.1 Defects in the body of a cast or forged, carbon or alloy steel valve may be repaired as permitted by the most nearly applicable ASTM cast or forged material specification listed in ASME B16.34.

6.3.2 The repair of defects in cast iron or ductile iron castings, by methods such as welding, brazing, plugging, or impregnation, is not permitted.

7 Marking Nameplates

7.1 API 594

Nameplates (see 5.8) of valves made in compliance with this standard shall be marked "API 594" and shall be additionally marked as follows:

a. For a valve made of steel, nickel alloy or special alloy, marking shall be in accordance with the requirements of ASME B16.34.
b. For a valve made of other materials, marking shall be in accordance with MSS-SP-25.

7.2 RESILIENT SEATING

Nameplates of valves furnished with resilient seating materials shall be marked to establish the temperature limitations of the seating material.

8 Shipment

8.1 COATINGS

8.1.1 The machined surfaces of materials that are not rust resistant shall be coated with an easily removable rust preventive.

8.1.2 Coatings shall not contain lead. Nonferrous and austenitic stainless steel valves shall not be coated. Unless otherwise specified in the purchase order, the external surfaces of valve bodies shall be painted with the following colors:

a. Carbon and low-alloy steel—aluminum.
b. Ductile iron—green.
c. Cast iron—black.

8.2 VALVE OPENINGS

8.2.1 Except for the ends of small, individually packaged valves, valve ends shall be covered to protect the gasket surfaces and valve internals during shipment and storage. The protective covers shall be wood, wood fiber, plastic, or metal and securely attached to the valve ends by bolts, steel straps, or suitable friction locking devices. The covers shall be no smaller than the outside diameter of the valve ends and designed so that the valves cannot be installed without complete removal of the covers.

8.2.2 All threaded connections in the valve body shall be fitted with solid, fully tightened plugs conforming to ASME B16.11, or ASME B16.14. Gray iron or malleable iron plugs shall only be used on gray iron or ductile iron valves respectively.

8.3 PACKAGING

8.3.1 When export packaging is not specified in the purchase order, valves may be shipped loose, palletized, or packed in a box or crate. Valves shall be packaged to prevent damage during shipment.

8.3.2 When the purchase order specifies export packaging, valves shall be shipped in wooden boxes or crates, individually or collectively, and packed to prevent their shifting within the package. (The shipping agent representing the purchaser will normally provide detailed instructions.)

9 Recommended Spare Parts

When specified on the purchase order, the vendor shall submit a complete list of spare parts. The list shall include cross-sectional or assembly-type drawings for identification with part numbers.

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